



CMPG325 INDIVIDUAL ASSIGNMENT

MILESTONE 1 - CLIENT DESIGN REVIEW.

PROJECT ID: CMPG325-2026-144

CLIENT ID: CL1-144

ASSIGNED ORGANISATION: LESIKA DATA ANALYTICS

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1.Client requirements:

Assigned organisation: Lesika Data Analytics (Potchefstroom)

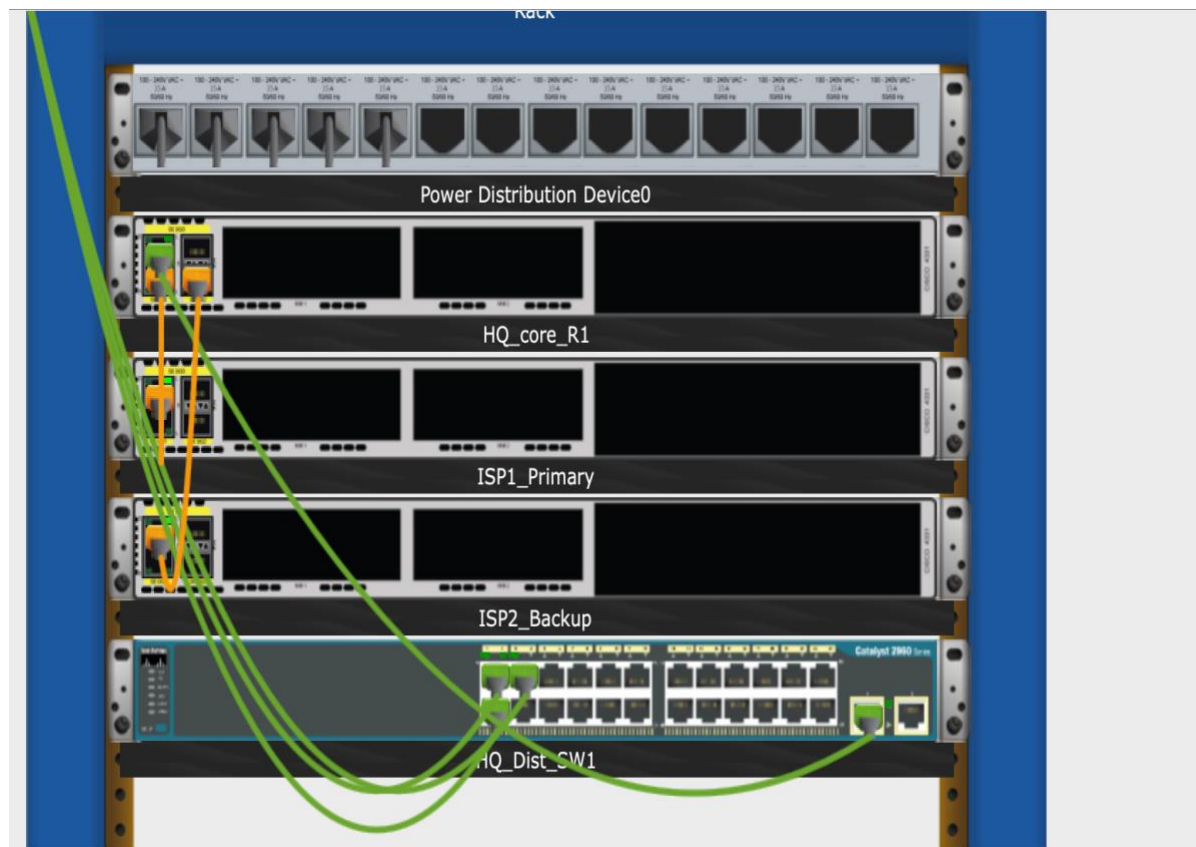
Industry: Technology

Main Goal/requirement:

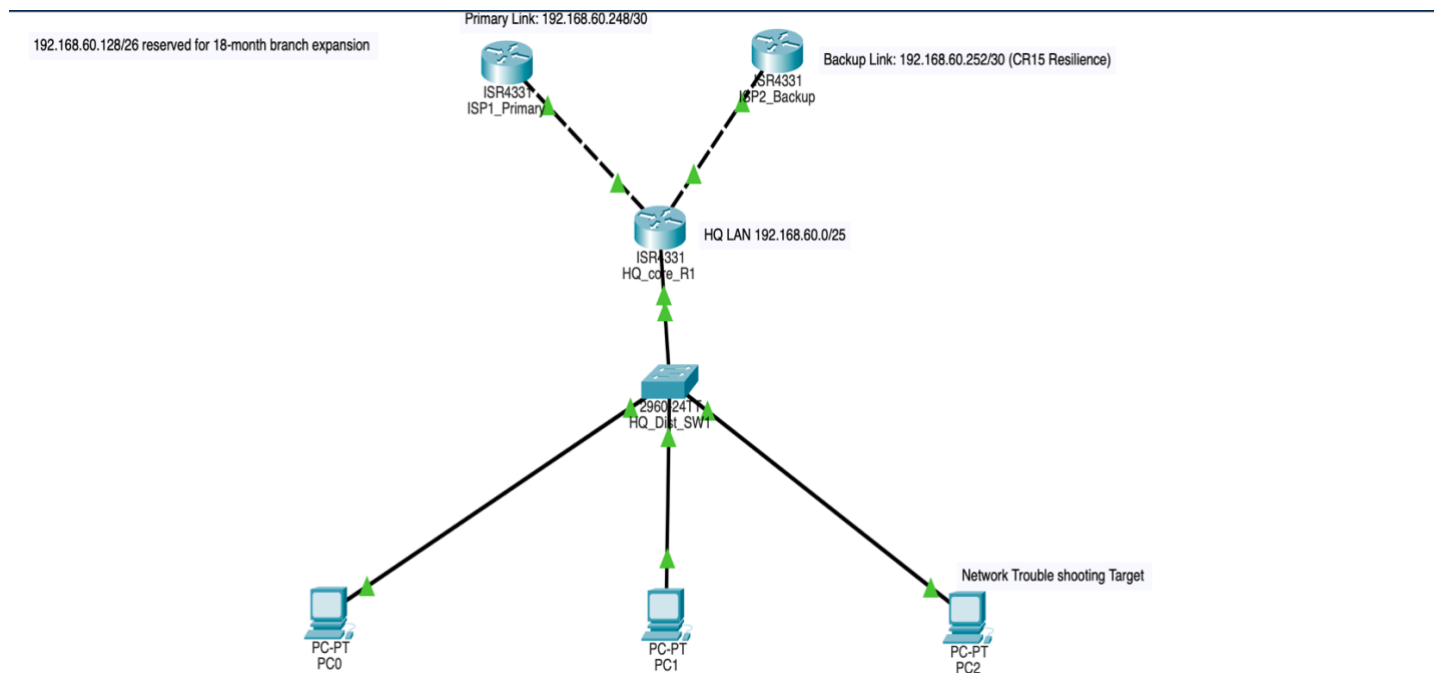
Design a network infrastructure that meets the following requirements:

- An infrastructure that is always available and online: That means they must use a dual-WAN design with a primary and a backup(FOR THE CR15 resilience requirement)
- Planned and organised IP address: The network IP addresses must be planned out and shared properly and correctly. Some addresses should also be saved for future use for a new branch opening in 18 months' time.
- LAN setup: The computers of the headquarters must be connected together in their own network(192.168.60.0/25).
- Troubleshooting: The network components must be labelled clearly so troubleshooting is easier.
- Documentation: The documentation of the network must show the logical and physical view of the network.

2. Physical topology:



3.Loggical topology:



4. IP Addressing plan:

Network Segment	Network Address	Subnet Mask	First Usable IP	Last Usable IP	Broadcast Address	Description
HQ LAN	192.168.60.0/25	255.255.255.128	192.168.60.1	192.168.60.126	192.168.60.127	Main Office
RESERVED	192.168.60.128/26	255.255.255.192	192.168.60.129	192.168.60.190	192.168.60.191	Reserved for branch Expansion
PRIMARY ISP	192.168.60.248/30	255.255.255.252	192.168.60.249	192.168.60.250	192.168.60.251	Point-to-point active link
BACK ISP	192.168.60.252/30	255.255.255.252	192.168.60.253	192.168.60.254	192.168.60.255	Point-to-point backup link

5. Initial GitHub repository:

https://github.com/KatDEv98/Final_Year_CMPG325_Networking_Project_Lesika_Data_Analytics_Network/upload/main